# Truth as a Compression Artifact
## in Language Model Training

NeurIPS 2026
Poster Session

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**Measured against *everything* the error generator can produce, the preference for the correct answer is small, late, and not about truth.**

The pairwise metric overstates it three- to fourfold; the coherent false rules of our main corpora are not filtered; a false selector the model can *learn* pushes the correct share below the null.

**Random errors, 86M**
$59.0 \pm 0.8\%$
mass on the correct completion; null 50; nothing below 86M

**One coherent rule**
$\approx 50\%$
main corpora, every size and architecture: not filtered

**Learnable false selector**
43%
correct share below the null against the family, 418M–3.08B

## 1. The Question

Real corpora contain contradictions: the same problem appears with a correct and an incorrect solution. Which one does a transformer trained from scratch put its probability on, when frequency, format and domain are held equal? Minimizing cross-entropy minimizes code length (Shannon 1948; Delétang et al. 2024), which suggests an error the model cannot predict from the problem is harder to accommodate than one it can. **We use this as motivation only**; both experiments built to make it a measured mechanism turned out to measure something else.

## 2. Measure the Whole False Side

The usual metric is pairwise: does the correct completion beat *one* incorrect one? If the error can take $N$ forms, a model that merely reproduces training frequencies puts 0.5 on the correct string and $0.5/N$ on each false one, and wins every pairwise comparison for $N \geq 2$.

We sum over every string the generator can produce instead:

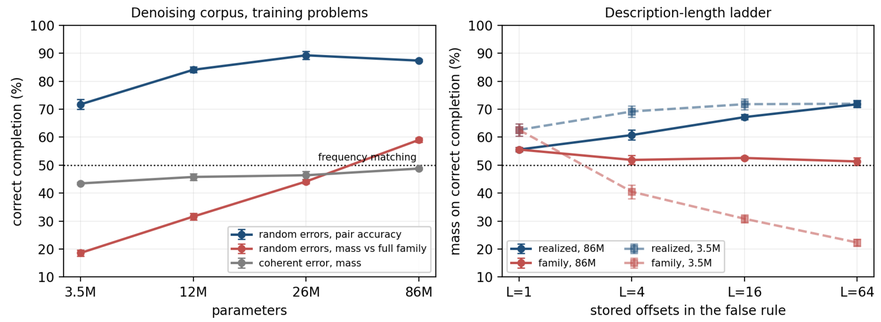
$$\text{share} = \frac{P(\text{correct})}{P(\text{correct}) + \sum_i P(\text{false}_i)}$$

Each problem occurs exactly twice in the primary corpus (once correct, once corrupted) and every other member of the family occurs zero times, so a frequency-matching model scores **exactly** 0.500 for any $N$. The share is conditional on the enumerated support; adding strings can only lower it, so it is conservative relative to the pairwise comparison. Full-string log-probability, not a per-token mean; a length-matched subset removes the remaining channel. Pair accuracy is reported only for comparability.

## 3. Experimental Design

GPT-2-like decoder-only transformers trained *from scratch* (3.5M–86M, 4 seeds) Qwen3-architecture models at 418M–3.08B (2–4 seeds). AdamW, lr $= 3 \cdot 10^{-4}$, 5,000 steps, character tokenizer. Four problem families (multi-step arithmetic, factoring, equation solving, differentiation) rendered as step-by-step derivations and verified with SymPy; only the *structure of the errors* varies: random per problem, one coherent false rule, $N$ competing rules, a rule with $L$ stored constants, or a false family whose selector the model can or cannot learn. Transfer: Wikipedia paragraphs with random or coherent entity substitution, and a 490M model on 1 GB of FineWeb-Edu with 8 % contradictory math. More than 500 training runs; per-run outputs released.

## 4. Central Contrast, and What the Pairwise Metric Adds




**Left:** denoising corpus, training problems, 4 seeds. Pair accuracy reads 72–89% at every size. Summing the generator's full support leaves an effect only at 86M: $59.0 \pm 0.8\%$ (every per-seed bootstrap interval above 0.5; 59.4% on an independent corpus). At 86M the realized-error score adds 27.6 pp of excess above chance and pair accuracy 37.4 pp where the family share gives 9.0 pp. The coherent rule sits at 43.5–48.8%; the shortfall is completion length ($-0.58$ tokens), not a preference.

**Right:** the description-length ladder (a single false rule storing $L$ constants). Against the one realized false string the share rises with $L$; against the full family it is flat at 86M (55.6, 51.9, 52.6, 51.3). **No grading in description length.**

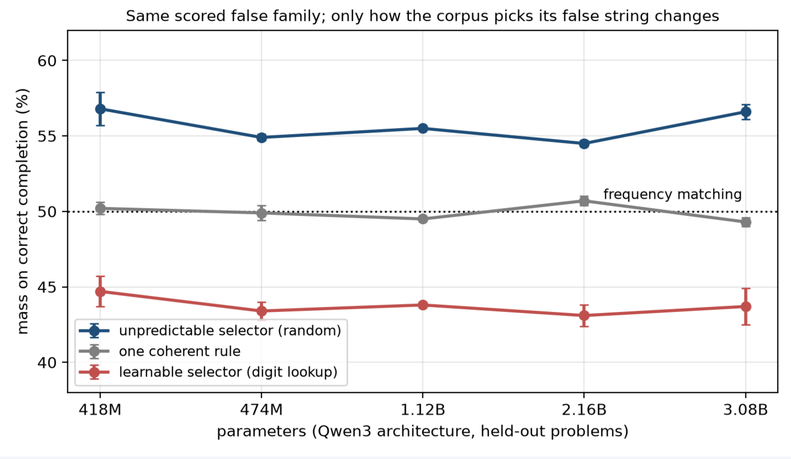
## 5. Two Results We Withdraw

**Multi-rule.** With $N$ competing false rules, pair accuracy jumps 47 → 79% at $N$=1 → 2 and reaches 90% at $N$=10. On the same checkpoints the aggregate share reads 47.9 → 47.8 → 38.7: the jump is *mass splitting*, not a selector cost. Reported as a calibration of how much pairwise signal splitting alone can manufacture.

**Description-length ladder.** Built to show a graded effect; the family measure is flat. A clean corpus with no false rule reads 99.7% at 86M and 64.1% at 3.5M on the same support, so the leak on a clean model is small where we report an effect and large where we report none.

Compression remains an *interpretation* of the surviving contrast, not a quantity these experiments measure.

## 6. What Moves the Share: Learnability




Held-out grid, evaluation problems disjoint from training, family share on the length-matched subset, null 50. Every selector is scored against the same random family (≈17.5 strings); the conditions differ in how the corpus picks its false string. Unpredictable selector: the correct system holds +5 points. Selector the model can learn (a visible digit): the share falls to $43.4 \pm 0.6\%$ at 474M (4 seeds), yet against the selected false string alone the correct completion still holds 65%; the deficit sits on the other family members. Partly learnable selectors sit at the null. The corpora share a generator but not their problems, so competence is not matched, and whether the below-null share is a shift toward the false system or lower confidence in the correct one is open. A clean corpus reads 96–99% at every size.

## 7. Transfer and Noise

**Wikipedia entity substitution** (3.5M–86M, 4 seeds, pair accuracy): a coherent global mapping ("France"→"Japan") reads 46–49% against a 50% null; a consistent substitution is not filtered on real text either. The random column (70–71%) has an unenumerable support and a null near 100%, and is not interpreted. **490M on FineWeb-Edu** with 8 % contradictory math: coherent $47.4 \pm 0.7\%$, not filtered (2 seeds).

**Noise ratio** on the controlled measure (86M, 2 seeds): with $K$ random errors per problem the null is $1/(K{+}1)$; the share reads 56.3, 44.9, 36.1 at $K$=1, 2, 4 against 50.0, 33.3, 20.0. The excess above the null does not shrink with noise.

## 8. Takeaway

*what this does and does not support*

- ► **What goes unfiltered is consistent or learnable structure, not falsehood as such.** The coherent rules of our main corpora keep their corpus share at every size (one ladder rung is the exception); a learnable false selector pushes the correct share below the null.
- ► **Pair accuracy is not a controlled diagnostic of preference.** Any design that varies the number of false forms (competing rules, noise ratios, entity substitution, corruption load) varies the metric with the treatment.
- ► **Not claimed:** compression as a measured mechanism, a graded description-length effect, an effect at every scale, or anything about production models.
- ► **For data curation:** diverse errors lose a little mass to the correct system; coordinated errors with a learnable rule were not filtered in our main corpora.